

The Research Agent

An AI-Powered Literature Review Workspace

Product Specification Document

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Core Vision

The Research Agent is a smart, dynamic, AI-powered literature review platform designed to liberate researchers from manual data entry. By combining PRISMA-compliant auditing, sentence-level source traceability, and automated report generation, the platform aims to accelerate thesis and systematic-review workflows by an order of magnitude.

1 Phase 1: Project Setup and Intelligent Discovery

1. Dashboard and Global Inventory

Upon logging in, users create a project. The dashboard presents a real-time status overview, including:

- Total papers read
- Pending / unread papers
- Total active clusters
- Top keywords, rendered as a list or word cloud

Recommendation: Add a lightweight “progress-to-goal” indicator (e.g., papers screened vs. target sample size) so the dashboard doubles as a PRISMA progress tracker, not just an inventory count.

2. Dynamic Clustering

Users can create custom clusters or categories tailored to their thesis topology. Clusters offer complete flexibility and can be added, renamed, or deleted at any time.

Recommendation: Support nested or tagged clusters (a paper belonging to more than one theme) rather than strict single-cluster membership, since real literature reviews rarely map cleanly to a single taxonomy branch.

2 Phase 2: PRISMA-Compliant Data Collection and AI Screening

3. BYOData and DOI Auto-Fetching

Users can drag-and-drop or browse their directories to upload multiple PDF files simultaneously. When a user inputs a DOI or URL, the system fetches metadata (title, abstract, year, citation count) via the Crossref or Semantic Scholar APIs. The system does not download the PDF automatically; the user must upload the file, and the system pairs it with the fetched metadata. A lightweight agentic AI routine assists by:

- Filling in metadata columns
- Assigning an initial cluster
- Suggesting keywords

Recommendation: Validate fetched metadata against the uploaded PDF's actual title/DOI on upload to catch mismatches early, and clearly flag any field the API could not resolve rather than leaving it blank.

4. AI PRISMA Screening (Inclusion / Exclusion)

Before papers enter the "Unassigned" queue, users define filtering rules in natural language (for example, "exclude papers published before 2020; exclude survey papers"). The AI reviews the corpus in seconds and routes non-compliant papers to an Excluded List. To strictly follow PRISMA 2020 guidelines, each exclusion displays either an exact supporting quote or a generated reason. Approved papers can be moved into clusters via drag-and-drop, drop-down selection, or a batch "Select to Transfer" menu.

Recommendation: Log every AI screening decision (rule applied, confidence, and evidence span) as an immutable audit record, and allow one-click reinstatement of excluded papers so reviewers can override the AI without losing the original PRISMA rationale.

3 Phase 3: The Smart Split-Screen Workspace (Deep Extraction)

5. Split-Screen Reading and Visual Extraction

Clicking a paper inside a cluster opens the Workspace. The left panel is a PDF viewer supporting zoom, pan, and highlighting; the right panel is a dynamic data-entry panel showing all columns, keyword tags, and edit controls, including quick actions to change cluster assignment or keywords. A bounding-box tool lets users draw a crop over diagrams, experimental tables, or equations; the selection is instantly saved as an image or extracted text in the corresponding cell.

Recommendation: Cache bounding-box crops at source resolution and store the page coordinates alongside the extracted asset, so figures can later be reused at full quality in the generated literature review report.

6. Dynamic Column Management and Split Logic

Users can create custom columns that automatically sync across every paper in a cluster. Columns can also be split into sub-columns — for example, dividing *Complexity* into *Time Complexity (TC)* and *Space Complexity (SC)*.

Recommendation: When a column is split, prompt the AI to re-derive values for the new sub-columns from the original column's existing evidence spans first, rather than re-scanning the full PDF, to save cost and preserve traceability continuity.

7. AI Auto-Fill and Human-in-the-Loop Verification

As soon as a user adds a new column, the Research Agent reads the full PDF and auto-fills the extracted values. AI-populated cells receive a yellow "AI Generated" badge; once a user reviews and confirms (or edits) the value, the badge turns green, marked "Human Verified."

Recommendation: Surface an aggregate verification rate per cluster (e.g., “62% human-verified”) so supervisors can gauge how much of the matrix is still AI-only before trusting a generated report.

4 Phase 4: Master Matrix and High-Scale Analysis

8. Cluster-Based Dynamic Spreadsheet

Returning to a cluster’s main page reveals a large, dynamic, Excel-like table capable of loading thousands of data points smoothly, with inline cell editing that avoids the need to reopen the split-screen view.

Recommendation: Virtualize row and column rendering from the outset (rather than as a later optimization) so performance stays smooth as clusters scale into the thousands of rows, and support column freezing for long-running reviews.

9. Sentence-Level Traceability (Source Citation)

Clicking any value in the Master Table or the right-side panel scrolls the left PDF viewer directly to the exact source; the specific sentence from which the AI extracted the data is highlighted in yellow.

Recommendation: Persist the sentence-level anchor as a stable offset (not a page-image coordinate), so traceability links survive PDF re-renders, annotation layers, or export to other formats.

5 Phase 5: Insights, Report Generation, and Sharing

10. Automated Unique Value Summary (Cluster Insights)

Beneath the table, or in a sidebar, the system analyzes column data and auto-renders a bulleted summary of unique values — for example, “Datasets Used: NSL-KDD, MNIST” or “Key Limitations: High memory usage, slow convergence.”

Recommendation: Let users pin or exclude specific values from the auto-summary, since raw uniqueness counts can be skewed by inconsistent phrasing (e.g., “MNIST” vs. “the MNIST dataset”) unless normalization is applied first.

11. AI Research Reports (Automated Thesis Drafts)

The “Generate Report” action prompts the AI to synthesize all papers in a cluster, together with the custom column splits and unique-value summaries, into a structured, 10–15 page literature review draft in paragraph form. Every claim carries an inline citation, and clicking a citation jumps directly to the corresponding line in the source PDF.

Recommendation: Generate the report section-by-section against a user-editable outline rather than as one long single pass, so a reviewer can regenerate or refine one subsection without discarding the rest of the draft.

12. Custom Export and Transparent Sharing

Users can export data by cluster, or select specific columns, as Excel (.xlsx), CSV, RIS, or DOCX files. A read-only or editable public link can be generated to share the entire split-screen dashboard with supervisors or co-authors, with no account required for the guest.

Recommendation: Give shared links configurable expiry and scope (e.g., single cluster only) by default, since an editable, no-login link to an entire project carries real data-exposure risk if it is ever forwarded beyond the intended recipient.

Business Model and Pricing Strategy (Draft)

Basic (Free) — Unlimited PDF uploads, basic column additions, manual data entry, and basic export.

Plus (\$15/mo) — AI PRISMA screening, column splitting, DOI auto-fetch, limited AI auto-fill.

Pro (\$40/mo) — Sentence-level traceability, table/figure bounding-box extraction, AI research report generation (10+ page drafts), unlimited AI extraction.

Enterprise (Custom) — University/lab integrations, unlimited team collaboration, and full API access.

Recommendation: Consider gating AI auto-fill and report generation by usage volume (tokens or pages processed) rather than purely by tier, since a single Pro user working across several large clusters can consume disproportionately more compute than the plan price anticipates.